INSY 8311 | FINAL EXAM | PHASE 0

SalonEase

Salon Appointment and Management System

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Database Development with PL/SQL

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GitHub: https://github.com/Bernice-hub12/bernice_27708_SalonEase



The notebook problem

Double bookings

Two customers booked with the same stylist at the same time.

Scattered records

Names, services and payments live on loose pages.

Unclear income

The owner cannot easily see daily or monthly earnings.

No insight

Nobody knows which services are popular or how many visits are cancelled.

Who is affected: receptionists, stylists, salon owners and their customers.

Why I chose SalonEase

From passion to a working system.



I love the salon world

Hair, braiding, makeup and nails are what I enjoy most.



A business I may start

I plan to open a salon business, so I want to understand how one runs and of course I'm planning to use the system instead of using papers manually.



It fits the course

It needs real database design, PL/SQL rules, triggers and reports.

Objectives

MAIN OBJECTIVE

Build an Oracle PL/SQL system that manages salon appointments, services and payments, prevents double booking and reports revenue.

SPECIFIC OBJECTIVES

1. Model the booking process (BPMN/UML swimlane)
2. Design a 3NF database loaded with real data
3. Write PL/SQL procedures, functions, cursors, a package and triggers
4. Build a user interface and dashboard on the live database
5. Record important changes in an audit log

Existing solutions and their gaps

EXISTING SYSTEMS

Fresha (hosted platform)

Online booking, SMS reminders, mobile app

L Six Hair Studio (Lim & Mat Nawi, 2023)

PHP web app, one studio, no measured results

Jajja Chinta salon (Fadil, 2025)

PHP e-appointment app; usability 4.03 / 5

GAPS SALONEASE FILLS

- Hosted platform: the salon cannot design its own database or rules
- Salon papers cover one business and show little measured evidence
- No-show research is on clinics (Nasir et al., 2020), not salons

Why it matters: hair and beauty businesses lose about 7% of monthly revenue to no-shows and cancellations (Fresha, 2026); beauty rescheduling rate 17.43% (Yocale, 2025).

Data source

1,952 x 22

Bookings x columns (Kaggle file)

11.5% no-shows

224 of 1,952 bookings

Toronto salon

Real data, Mar to Jul 2018

SAMPLE RECORDS (stylist names masked)

book_dow	book_category	book_staff	last_receipt_tot	recency	noshow
Wednesday	STYLE	Stylist 1	262	35	0
Saturday	STYLE	Stylist 3	35	9	0
Friday	COLOR	Stylist 2	55	13	0
Saturday	STYLE	Stylist 3	100.4	15	1

Supplement: a simple survey of Kigali salons for services and prices, plus clearly labelled generated data for clients, dates and payments.

Licence: CC0: Public Domain| Link: https://www.kaggle.com/datasets/frederickferguson/hair-salon-no-show-data-set?

Proposed solution and innovation



User interface

- Register clients
- Book appointments
- Manage services
- Record payments



Oracle database

- PL/SQL business rules
- Blocks double booking
- Calculates totals
- Audit log



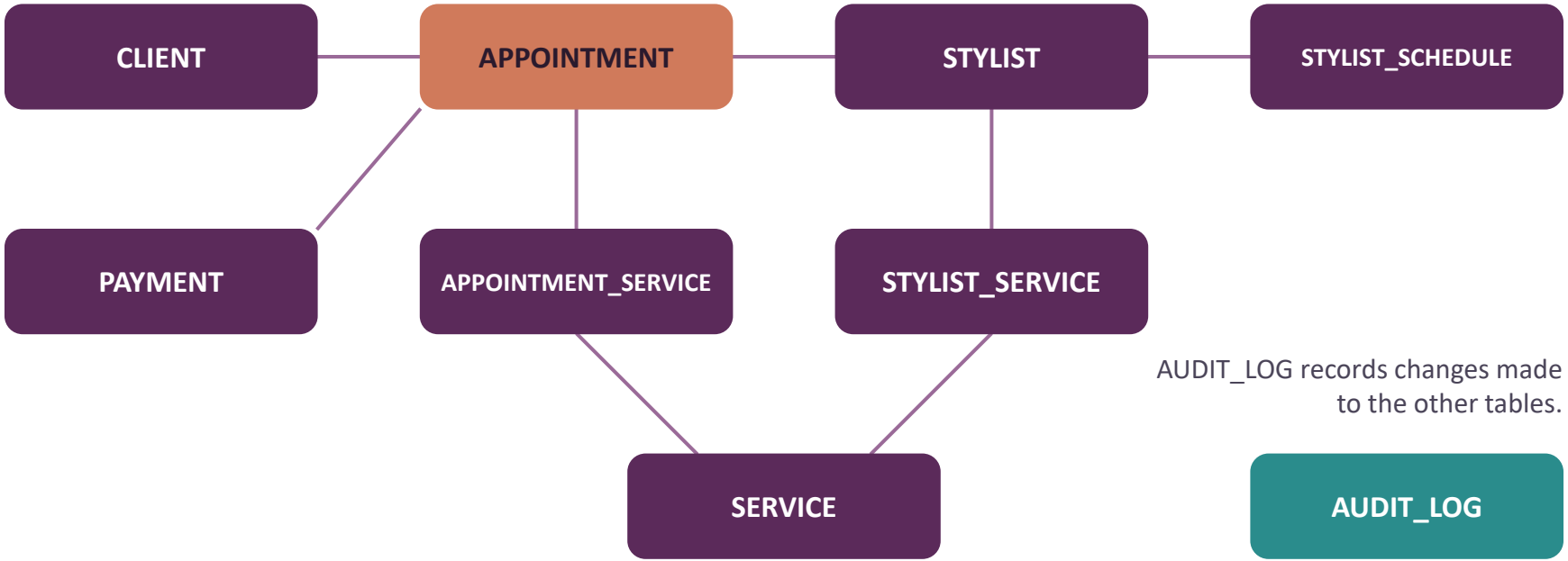
Dashboard

- Daily bookings
- Monthly revenue
- Appointment statistics
- Popular services

Innovation option: User interface + Analytics dashboard
Tool: Oracle APEX on the live Oracle database

Planned database

A client has many appointments; each appointment has one stylist, many services and its payments.



PL/SQL approach: where each feature is used

Procedure book_appointment	Saves a booking only after checking the stylist is free
Procedure record_payment	Records payment after the service is completed
Function is_stylist_available	Checks the stylist is free at the chosen time
Function calc_appointment_total	Adds the prices of all services in one visit
Cursors	Loop over completed visits for daily and monthly revenue
Package salonease_pkg	Groups booking, payment and report logic
Trigger: double booking	Blocks two appointments for one stylist at once
Trigger: audit log	Logs who changed or cancelled an appointment
Restriction rule	Blocks bookings in the past or after opening hours
Exceptions	Show a clear message when a stylist is not available

Expected results: the live demo

1

Book

Create an appointment in the app

2

Block

A double booking is rejected

3

Pay

Record payment after the service

4

See

Dashboard shows revenue and popular services

5

Audit

Changes appear in the audit log

Limitations and future improvements



Limitations

- Main dataset comes from one Toronto salon
- Some tables use survey or generated data
- Staff-only system, customers cannot book online



Future improvements

- SMS or email reminders
- Online booking for customers
- No-show prediction from the dataset
- Stock and product management

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Thank you. Questions?